1. Experiment in short

Number of nodes	5
Number of topologies	1
Location of the experiment	Hadera, Israel $(32.824165°\text{N}, 35.000870°\text{E})$
Duration of each measurement	600 seconds
Link sampling interval	20 second
Link quality measurements	SNR

2. Location

The experiment took place in May 2017 in the vicinity of the Hadera coal pier, Israel. The water depth is 25 m. The pier's stands are made of hard concrete, and the pier is roughly 15 m above the sea level. The experiment was performed in harsh conditions with waves approaching 1.5 m of height. Moreover, the pier environment included acoustic noise of time-varying level caused by harbor activity. This yielded a large variance for the signal to noise ratio (SNR), which ranged between 10 dB and 40 dB. A n illustration of the experiment setup is shown in Fig. 1.

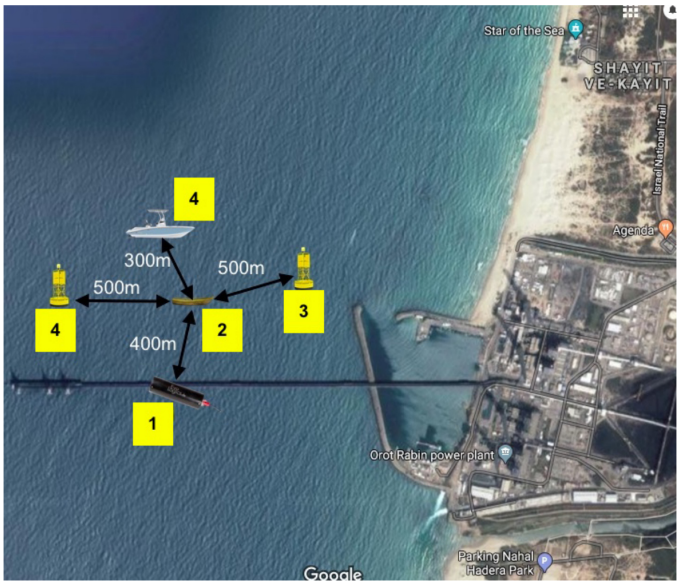


Fig. 1. Illustration of the sea trial location (picture taken from Google maps on June 3, 2018). The different locations of the vessels are marked.

3. Topologies

The network included five nodes. Node 1 was stationed on the pier; Nodes 2 and 5 were represented by two free floating small vessels; Nodes 3 and 4 were moored acoustic receivers with a short rope that

allowed location shift of up to 10 m. Only Nodes 1 and 5 were transmitting, while Nodes 2, 3, and 4 were receiving. All transmitter-receiver links were operational.

4. Description of the nodes

For transmission and reception, Nodes 1, 2, and 5 used a EvoLogics S2CR 7-17 software-defined modems operating in the 7 kHz-17 kHz band. The modems were connected to an external source and to a laptop computer via a 30-m Ethernet cable. All three nodes deployed their modems at a depth of 10 m. For reception, Node 2 transferred a sequence of buffers to a connected laptop computer, while Nodes 3 and 4 used a Cetacean uRadar acoustic self-recorder. The latter uses an internal recorder suitable for frequencies up to 48 kHz, and the recorders were also deployed at depth of 10 m. All three receivers were self-synchronized based on the transmission times from Nodes 1 and 5.

In each transmission, Nodes 1 and 5 emitted bursts of linear chirps spanning the full 7-17 kHz bandwidth. The symbols' duration was 90 ms, and a guard interval of 10 ms was used. Transmissions were performed with a source level of 165 dB Re 1μPa@1m. Node 1 made 7 burst transmissions, and Node 5 made 3 burst transmissions. The received signals were analyzed offline to extract channel features.

5. Tested protocols and schemes

The purpose of this experiment was to test a physical layer security algorithm, and thus no communication packets were used. Instead, Nodes 1 and 5 transmitted 7 and 3 bursts of 100 symbols, respectively, each spaced by a delay of 1 minute.

6. Link quality measurements

The quality of the network links is evaluated by means of the SNR. The SNR was measured at the output of the matched filter for each received symbol, and an average for each burst is used. The result is a time-varying, per-link SNR.

7. Available data

The stored data is given in the `HaderaSecurity-05-17.mat` file. The file contains four types of matrices:

- the location matrix `LocMat`, of size $9\times5\times3$;
- the topology measurement matrix `TopMat`, of size $9\times5\times5$;
- the adjacency matrix `AdjMat` of size $9\times5\times5$;
- the technology matrix `TechMat` of size $9\times5\times1$.

Using Matlab/Octave matrix notation:

- `LocMat(t,i,:)` are the UTM coordinates (zone 36S) and the depth of node i at time instant t;
- `TopMat(t,i,j)` is the SNR for link $i \rightarrow j$ at time t;
- `AdjMat(t,i,j)` is equal to 1 if there exists a link between nodes i and j at time t;
- `TechMat(t,i,p)` is equal to 1 if node i can use physical layer technology p at time t.

8. Reference to detailed description

Further details about this experiment are provided in the following paper:

- R. Diamant, P. Casari, and S. Tomasin, "Cooperative Authentication in Underwater Acoustic Sensor Networks," *IEEE Transactions on Wireless Communications*, vol. 18, no. 2, pp. 940–949, Feb. 2019.